APPS Platform - Muzhu Features & Usage Manual

AI Predictive Personnel Stress & Welfare Monitoring System — Built for Defense Forces & IT Engineering Teams

Indha official technical document-la namma APPS system start panna naal-la irundhu ippo varaikkum build panna **ellaa features, tools, clinical algorithms, security mechanisms, matrum UI components** patri detailed-ah Tanglish-la vilakkapattulladhu. Indha platform Defense Armed Forces matrum IT / Software Engineering / Corporate high-stress teams-kaga dual-domain architecture-la design panna pattu irukku. Soldiers field stress matrum IT developers sprint burnout, night shift fatigue, continuous on-call exhaustion-ai early stage-laye predict panni prevent seiyum modern solution. Ovvoru feature-um epadi work aagudhu, adhai epadi use pannanum, yaaru use pannanum enbadhai keezhe detailed-ah paarkkalaam.

Document Title:	APPS Complete System Features & User Manual (Tanglish)
Target Audience:	Defense Service Members, IT Software Engineers, Tech Leads, Welfare Officers, Engineering Directors & Admins
System Core Engine:	Python Flask, SQLite3, Vercel Blob Persistent Snapshot Engine, CSS3 Flex/Grid
Security Baseline:	PBKDF2 Password Salting, Command PIN Gating, Immutable Activity Audit Trail
Live Production URL:	https://apps-nu-henna.vercel.app (Deployed on Vercel)
Online PDF Viewer:	/guide (Features Guide) • /documentation (System Docs)
Repository:	https://github.com/dharuntech275/APPS- (Dual tracked: main & master)

1. System Overview & 3-Tier Command Architecture

APPS platform armed forces jawan-galukku matrum welfare officers-kku oru reliable digital bridge-ah vela seiyudhu. System-la motham **3 mukkiya roles** irukku:

- **1. Personnel (Soldier / Jawan):** Dhinamum thanga stress, thoongum neram (sleep hours), heart rate, matrum family morale status-ai log panni thanga mental fitness-ai track pannalaam. Instant panic relief-ku 4-7-8 breathing exercise matrum emergency 1-click Crisis SOS call seiyalaam.
- **2. Welfare Officer:** Thanga command unit-la irukkum anaithu jawans-oda health status-ai live-ah monitor pannalaam. High-risk red alert jawans-ai prioritize panni clinical counseling notes add panni suicide prevention actions edukalaam.
- **3. Senior Admin (HQ Command):** Unit-level statistics (High, Medium, Low risk counts), comprehensive risk analytics, audit trail activity logs, matrum overall system oversight control seiyalaam.

Mukkiya Kurippu: Starting-la login issue irundhadhu, adhai streamline panni login direct aakkiyom. Pinnar 2FA shield kooda thevai illai enbadhaal UI, database, matrum backend flow-la irundhu clean-ah remove panni friction-free direct login ready panniyom.

2. Authentication, Account Management & Security Features

2.1 Direct Secure Login & Session Management

Enna Feature: User thanga username matrum password enter panni direct-ah thanga dashboard-kku login seiyyum system. 2FA TOTP shield thevai illai endru sonnadhukku apparam, 6-digit authenticator code thevai illamal direct single-step login create panni irukkom.
Epadi Work Aagudhu: User credentials enter pannumpodhu, backend-la SQLite database query panni password verify aagum. Password match aana udane HTTP-only permanent signed session create aagi, user-oda role-kku thagundha dashboard-kku redirect aagum (`/dashboard`, `/officer/personnel`, or `/admin_dashboard`).
Benefit: Field-la network speed kuraivaaga irukkum podhum jawan-gal friction illamal 2 seconds-la login panni daily log submit panna mudiyum.

2.2 Frictionless Multi-Role Direct Registration

Enna Feature: User request-kku inanga Command Security PIN-ai neekki, Personnel, Welfare Officer, matrum Senior Admin aana yaarum direct-ah thanga role-ai select panni register seiyyum instant onboarding system.
Epadi Work Aagudhu: Signup screen-la 'Personnel', 'Officer', alladhu 'Admin' tab-ai click panni, Name, Unit, and Password enter panninaale podhum. Zero delay-la account create aagi respective dashboard-kku login panna mudiyum.
Benefit: Onboarding speed adhigam aagividum, network illaatha idathilum staff PIN authentication error illamal account create seiyalaam.

2.3 Cryptographic PBKDF2 Password Hashing & Weak Password Upgrade

Enna Feature: Security upgrade feature. Pazhaiya plain-text alladhu weak passwords-ai defense-grade PBKDF2-HMAC-SHA256 salting-kku automatic-ah upgrade seiyyum mechanism.
Epadi Work Aagudhu: User login pannumpodhu password weak-aga irundhaalo alladhu unhashed-aga irundhaalo, system udane `/upgrade_password` modal popup kaattum. User 8+ characters strong password set panniyavudan cryptographically hash aagi database-la update aagividum.
Benefit: Military staff accounts brute-force attacks matrum credential theft-la irundhu safe-ah irukkum.

2.4 Self-Service Password Recovery (/forgot_password)

Enna Feature: Soldier password marandhu poitaal, external email alladhu SMS OTP thevai illamal thanga unit verification moolam password-ai reset pannum defense self-service tool.
Epadi Work Aagudhu: Login screen-la 'Forgot Password?' click panni, Registered Name matrum Service Unit (e.g. 15 Rajputana Rifles) submit pannanum. Record match aana udane puthiya password set panna allow pannum.
Benefit: Remote border outpost-la phone network / OTP service illadhavargalukku indha unit-verified recovery migavum help-ah irukku.

2.5 Login Page Streamlining & Cleanup

Enna Feature: Neenga ketta maadhiri login page-la irundha thevai illadha banners, unnecessary test links, matrum clutter-ai remove panni clean executive login card-ah maathiyadhu.
Epadi Work Aagudhu: Clean card layout, direct username/password fields, role-based registration link, matrum emergency hotline reference mattum display aagum.
Benefit: User confuse aagaamal fast-ah login seiyalaam.

3. Personnel (Soldier / Jawan) Features & Detailed Usage

3.1 Daily Mental Health & Wellness Assessment Survey

Enna Feature: Jawan thanga daily psychological and physical health telemetry-ai submit pannum core assessment form. Idhula 4 clinical metrics irukku:
• **1. Sleep Duration (Hours):** 0 mudhal 12 mani neram varai thoongiya alavu.
• **2. Operational Stress Level (1 - 10 Slider):** Interactive slider. Move pannumpodhu Dynamic Qualitative Badges marum: 1-3 (Calm & Focused - Green), 4-5 (Manageable Duty Load - Slate), 6-7 (Elevated Tension - Amber), 8-10 (Critical Combat Stress / High Risk - Red).
• **3. Heart Rate / Pulse (BPM):** 40 mudhal 180 bpm varai physical exertion alavu.
• **4. Social / Family Morale Connection (1 - 5 Stars):** Family and squad support level.
• **Field Operational Notes:** Thangalukku irukkum bayam, thoongamai, alladhu duty shramathai pathi ezhuthalaam.
Epadi Use Pannanum: Personnel dashboard-kku vandhu sliders-ai adjust panni, 'Log Daily Wellness Entry' button-ai click pannanum.
Benefit: Jawan-oda daily mental status continuous-ah track aagum.

3.2 Automated Psychological Triage Scoring Algorithm

Enna Feature: Jawan log submit panna udane, AI clinical heuristic algorithm automatic-ah triage level-ai calculate pannum:
• **High Risk (Urgent Action Required - Red):** Sleep < 5 hours, Stress >= 8, Heart Rate > 100 bpm, alladhu Social Connection == 1.
• **Medium Risk (Moderate Caution - Amber):** Sleep 5-6 hours, Stress 5-7, alladhu Heart Rate 90-100 bpm.
• **Low Risk (Stable Mental State - Green):** Adequate sleep (>= 7h), Stress < 5, normal heart rate.
Benefit: Welfare officers immediate-ah yaara contact pannanum-nu automated-ah theriyum.

3.3 Tactical 4-7-8 Breathing & Relaxation Stabilizer

Enna Feature: Jawan-galukku panic attack, battle stress, alladhu sleeplessness varumpodhu parasympathetic nervous system-ai activate seiyyum breathing tool.
Epadi Work Aagudhu: Dashboard-la '4-7-8 Relax' quick action button alladhu full card irukku. 'Start Session' click panninaal:
1. **Inhale (4s):** Animated circle expand aagum, mooku valiyaaga moochu ulle edukka vendum.
2. **Hold (7s):** Circle steady-ah irukkum, moochai adakki vaikka vendum.
3. **Exhale (8s):** Circle shrink aagi moochai vaai valiyaaga methuvaaga veliye vida vendum.
Dynamic cycle counter (Cycle 1 of 4) matrum animated progress bar display aagum.
Benefit: 2 minutes-kulle heart rate kuraithu instant mental calmness kidaikkum.

3.4 Simulated Smartwatch / BLE Telemetry Sync

Enna Feature: Wearable fitness bands alladhu army smartwatches (BLE) illadha idathil kooda, telemetry sensor-ai simulate panni single click-la realistic sleep and heart rate data-vai fill pannum tool.
Epadi Use Pannanum: Daily survey form-la irukkum 'Sync Smartwatch Data' button-ai click seiyyum podhu, realistic military biometrics (Sleep: 6.8h, Heart Rate: 72 bpm) automatic-ah form fields-la load aagividum.
Benefit: Form fill seiyyum neram kuraivadhaal jawans regular-ah entry poduvargal.

3.5 1-Click Crisis SOS Emergency Beacon

Enna Feature: Soldier extreme mental breakdown, suicidal thoughts, alladhu severe panic situation-la irukkumpodhu 1-click-la emergency help kidaikkum defense beacon.
Epadi Work Aagudhu: Header-lum dashboard banner-lum prominent RED SOS button irukku. Click panninaal:
1. Army Emergency Helpline (**1902**) & National Tele-MANAS (**14416**) direct call modal open aagum.
2. Welfare Officer matrum Admin screen-la urgent red pulsing banner alert dispatch aagum.
Benefit: Life-saving crisis intervention immediate-ah kidaikkum.

3.6 Cumulative Fatigue Forecaster & 7-Day Sleep Debt Engine

Enna Feature: Soldier kadandha 7 naatkalaga evvalavu thookkam izhanthirukirar (sleep debt) enbadhai calculate seiyyum clinical algorithm.
Epadi Work Aagudhu: Baseline military requirement 7 hours daily sleep. Daily sleep deficit-ai kooti **Operational Strain Index (0 - 100%)** calculate seiyudhu. Strain index 65%-kku mela ponaal, 'Severe Sleep Debt Alert - Mandatory Rest Required' endru recommendation kaattum.
Benefit: Prolonged fatigue aala varum weapon handling accidents matrum guard duty lapses-ai thadukkum.

3.7 Gamified Morale Badges & Daily Streak Tracking

Enna Feature: Jawans thodarndhu daily entry pottu thanga health-ai track panna motivate pannum gamification system.
Epadi Work Aagudhu: Consecutive days entry pottaal streak increase aagum (e.g. 5-Day Streak ■). Athil 'Mental Resilience Sentinel', 'Sleep Master', 'Iron Mind' pontra military achievement badges unlock aagum.
Benefit: Mental health entry oru duty-ah feel aagaamal positive habit-ah maarrum.

3.8 Soldier Personal Health Dossier PDF Export (/my/export_pdf)

Enna Feature: Jawan thanga mudiyatha health history, past surveys, triage levels, matrum fatigue index-ai professional medical PDF report-ah download seiyyum feature.
Benefit: Base medical officer-kku physical check-up panna poga podhu indha PDF report-ai katti consultation edukalaam.

4. Welfare Officer Command Suite & Clinical Management

4.1 Personnel Live Monitoring Roster & Triage Filter

Enna Feature: Officer thanga command unit-la irukkum anaithu jawans-oda current psychological triage status-ai live-ah paarkkum central dashboard.
Epadi Work Aagudhu: Personnel list table-la ovvoru jawan-oda Name, Service Unit, Latest Sleep, Stress Level, Heart Rate, Triage Tag (High/Medium/Low), matrum Action link display aagum. Search bar matrum filter moolam High Risk jawans-ai mattum instant-ah isolate panni paarkkalaam.
Benefit: 100+ jawans irundhalum, risk-la irukkum 3 jawan-ai 2 seconds-la identify panna mudiyum.

4.2 Soldier Longitudinal Clinical Dossier

Enna Feature: Oru specific jawan-oda complete past records, date-wise mental trends, sleep debt graph, matrum biometric changes-ai kaattum detailed profile.
Epadi Use Pannanum: Personnel list-la jawan name pakkathula 'View Dossier' click seiyya vendum.
Benefit: Sudden mood drop alladhu prolonged combat fatigue eppo start aanadhu endru accurate-ah track panna mudiyum.

4.3 Confidential Clinical Counseling Notes Engine

Enna Feature: Welfare Officer jawan-kku 1-on-1 counseling kuduthu vittu, thanga clinical findings, action steps, matrum next review date-ai confidential-ah log pannum tool.
Epadi Work Aagudhu: Dossier page-la 'Add Clinical Counseling Note' form irukku. Officer note type (General Counseling, PTSD Intervention, Family Stress) select panni text submit panninaal, athu encrypted SQLite `CounselingNotes` table-la save aagum matrum `ActivityLog`-lum immutable audit entry vizhum.
Benefit: Medically structured record maintain aagum; soldier transfer aanaalum new officer-kku past background theriyum.

4.4 Emergency SOS Alert Banner Intercept

Enna Feature: Jawan SOS button-ai trigger panninaal, officer-oda screen-la top position-la pulsing crimson RED banner display aagum.
Benefit: Officer vera edhavathu page-la irundhaalum, emergency case-ai miss panna maattaargal.

4.5 Unit Welfare Assessment Report PDF Export (/officer/export_report)

Enna Feature: Motha unit-oda triage distribution, high-risk case summary, matrum counseling follow-up details-ai formal military briefing PDF document-ah export seiyyum tool.
Benefit: Commanding Brigade alladhu Battalion Headquarters-kku weekly welfare status report submit panna idhu direct-ah use aagum.

5. Senior Admin Command Headquarters & Operations

5.1 Real-Time Executive Command Analytics Dashboard

Enna Feature: Headquarters Admin-kku armed forces units-oda overall mental readiness-ai bird's-eye view-la kaattum analytics overview.
Epadi Work Aagudhu: Total Registered Personnel count, High Risk Soldiers count (Red Card), Medium Risk count (Yellow Card), and Stable Low Risk count (Green Card) live metric cards-ah display aagum. Unit-wise summary table-la ovvoru regiment/battalion-la evvalavu per risk-la irukkanga endra data varum.
Benefit: Top brass commanders endha unit-la operational stress adhigama irukku-nu therindhu squad rotation alladhu leave sanction seiyalaam.

5.2 Mobile Responsive Drawer & Touch Layout System

Enna Feature: Smartphone-la view pannum podhu elements olunga theriyadha prachanai-ai fix panna build panna mobile-first responsive layout.
Epadi Use Pannanum: Mobile phone-la open pannumpodhu slide-out drawer menu, side-by-side equal height quick action buttons (`Crisis SOS` and `4-7-8 Relax`), 5-column mood selector, and zero horizontal scroll automatically fit aagum.
Benefit: Field-la chinna screen phone-la use pannaalum perfect alignment and touch accessibility kidaikkum.

5.3 Immutable Operational Audit Trail (ActivityLog)

Enna Feature: System-la nadakkum anaithu critical actions-aiyum timestamp, actor ID, target soldier, matrum action type-udan record pannum defense audit trail.
Epadi Work Aagudhu: Login, counseling note entry, PIN change, triage status transition ellaame append-only table-la save aagum. Yaarume indha logs-ai edit alladhu delete panna mudiyadhu.
Benefit: Military security compliance matrum complete operational accountability urudhi aagum.

5.4 Senior Command Summary PDF Export (/admin/export_report)

Enna Feature: Top-level military executive report ready pannum generator.
Benefit: Monthly readiness briefings-kku presentation-ready formatted report kidaikkum.

6. UI/UX Layout, Localization & Cloud Infrastructure

6.1 Multi-Language Vernacular Switcher (English, ■■■■■, ■■■■■)

Enna Feature: Soldier thanga thaa mozhiyil (Tamil, Hindi, alladhu English) system-ai use seiyyum client-side instant language toggle.
Epadi Work Aagudhu: Header-la dropdown select box irukku. Language choose panninaal zero-latency-la page-la irukkum ellaa text-um instant-ah switch aagum. Indha preference browser-oda `localStorage`-la save aavadhaal, adutha murai open pannumpothum avargalukku thaa mozhiyileye varum.
Benefit: English theriyadha jawans kooda easy-ah survey fill panni mental health assistance pera mudiyum.

6.2 Laptop Executive 2-Column Grid & Mobile Responsive Drawer Layout

Enna Feature: Neenga ketta laptop view and mobile alignment request-kku yetharpola build panna responsive design system.
Epadi Work Aagudhu:
• **Laptop Screens (>= 992px):** Modern 2-column executive grid (`1.15fr 0.85fr`). Left column-la survey card, right column-la fatigue forecaster matrum gamified morale badges pakkam pakkamaaga clean-ah render aagum. Horizontal scroll illamal compact executive presentation kidaikkum.
• **Mobile Screens (< 768px):** Touch-friendly single column vertical flow. Navigation bar hamburger icon moolam slide-out drawer menu-ah maarrum. Quick action buttons (`Crisis SOS` and `4-7-8 Relax`) side-by-side equal height wrap aagi touch panna comfortable-ah irukkum. Page-la horizontal overflow (`overflow-x: hidden`) completely zero aakki irukkom.
Benefit: Soldier smartphone-la use panninaalum, Welfare Officer laptop-la use panninaalum flawless layout kidaikkum.

6.3 Vercel Serverless Architecture & Private Vercel Blob Persistence Engine

Enna Feature: Cloud-la deployment panna podhu, serverless container shut down aanaalum database alindhu pogamal thadukkum zero-data-loss synchronization engine.
Epadi Work Aagudhu: Vercel-la standard container ephemeral (temporary) aagum. Idhai solve panna custom `sync_db_to_blob()` matrum `sync_db_from_blob()` engine build panniyom. Ovvoru murai survey, note, alladhu PIN update pannumpothum, backend database SQLite file snapshot private Vercel Blob store (`apps-data` store ID: `store_eS9r7map43Dx32yX`)-kku instant-ah upload aagum. Pudhiya container wake up aagumpodhu blob store-la irundhu restored database automatic-ah hydrate aagividum.
Benefit: Free cloud hosting-la kooda 100% persistent defense database guarantee kidaikkudhu.

6.4 2FA Shield Removal & Login Streamlining Refactor

Enna Feature: Starting-la build panna Google Authenticator (RFC 6238 TOTP) shield-ai, user request-kku inanga complete-ah remove panniyom.
Enna Maathinom:
1. Desktop header matrum mobile drawer-la irundha '2FA Shield' links anaithum neekkiyom.
2. Admin Dashboard-la irundha '2FA Security Roster Card' matrum reset buttons neekkiyom.
3. Login challenge intercept (`/login/2fa`) remove panni direct single-step login create panniyom.
4. Database-la irundha `totp_enabled` flags reset panni Vercel Blob cloud store-lum sync panniyom.
Benefit: Login process speed adhigamaagi, zero unnecessary verification step-la soldiers access panna mudiyudhu.

6.5 Dedicated Header [■ Menu] Capsule Navigation & Vector SVG Icons

Enna Feature: User request-kku inanga Option 1 Dedicated Menu Capsule Button (`#headerMenuBtn`) matrum 100% sharp high-DPI Vector SVG icons system build panniyom. Pazhaiya informal emojis anaithum complete-ah neekki, military & enterprise-grade clean iconography-ah maathiyom.
Epadi Work Aagudhu:
• **Dedicated Menu Pill:** Top command header-la neat-ana 4-square vector grid icon + 'Menu' text-udan tactical capsule button amaikkappattulladhu. Adhan pakkathil irundha duplicate 'APPS' text matrum brand shield tick icon anaithu 3 roles (Personnel, Officer, Admin)-lum neekkappattu, Menu button mattum highlight aagum.
• **Instant Touch & Click Execution:** Inline duplicate event conflicts solve panni, single authoritative menuBtn.onclick handler bind seiyappattulladhu. Drawer z-index 9999-ku elevate panni Laptop matrum Mobile rendulayume 0.28s smooth slide-in navigation open aagum.
• **Zero Informal Emojis:** Platform-la irundha emojis anaithum SVG vector paths-aga redesign panni Retina displays-lum sharp clarity tharum.
Benefit: Mobile matrum laptop screens-la navigation instant-ah open aagi, modern military & SaaS standard look kidaikkum.

7. Multi-Industry Adaptability: Defense Forces & IT / Software Engineering Applications

APPS platform armed forces military regiments-kku mattum illaamal, **IT companies, software engineering teams, SRE squads, matrum high-stress corporate enterprises-kum** direct-ah use aagum vagaiyil expand seiyappattulladhu. High-stress environments-la irukkum workers psychological burnout-ai early-stage-la track seiya idhu universal solution-ah vela seiyudhu.

7.1 Defense vs IT / Enterprise Role & Metric Equivalence

System Core Dimension	Armed Forces / Defense Use Case	IT & Software Engineering Enterprise Use Case
Personnel Role	Combat Soldier, Field Jawan, Border Guard	Software Developer, DevOps Engineer, 24/7 SRE on-call Engineer
Welfare Officer Role	Regimental Welfare Officer, Battalion Medical Officer	Engineering Manager, Tech Lead, Scrum Master, People Ops / HR Lead
Admin Role	HQ Unit Commander, Brigade General Staff	VP of Engineering, CTO, Chief People Officer, HR Director
Duty Hours Metric	Patrol shifts, sentry guard duty, field deployment hours	Sprint coding hours, emergency on-call incident shifts, crunch periods
Operational Stress (1-10)	Tactical operations tension, isolation, high-altitude fatigue	Production outage downtime stress, sprint delivery pressure, merge blockers
Smartwatch Telemetry Sync	Garmin Instinct / Military tactical wearable biometrics	Apple Watch, Fitbit, Samsung Galaxy Watch, WearOS band biometrics
4-7-8 Breathing Tool	Post-patrol combat decompression, adrenaline stabilizer	High-severity production P0 post-mortem decompression, anxiety relief
APPS AI Wellness Companion	Confidential psychological decompression for remote jawans	24/7 empathetic assistant for developer burnout, insomnia & shift fatigue
Emergency SOS Beacon	Direct distress signal to Battalion Welfare Officer & Army 1902 helpline	Instant confidential escalation to People Ops Lead & Employee Assistance (EAP)

7.2 Direct Browser PDF Viewer Endpoints (/guide & /documentation)

Enna Feature: Documentation matrum Features Guide-ai download pannaamal direct-ah browser tab-le open panni padikkum vasadhi.
URLs:
• https://apps-nu-henna.vercel.app/guide — Opens indha Features Guide PDF directly in the browser.
• https://apps-nu-henna.vercel.app/documentation — Opens the System Technical Documentation PDF directly in the browser.
• Navigation Drawer matrum Landing Portal-la direct 1-click links include seiyappattulladhu.
Benefit: Users endha device-la irundhaalum (Mobile / Laptop) PDF viewer app thevai illamal browser-le manual-ai padikkalaam.

8. Comprehensive Feature Matrix & Usage Cheatsheet

Keezhe kanda table-la namma platform-oda anaithu features, athai yar yaaru use pannalaam, matrum adhan primary use case quick reference summary-ah kudukkappattulladhu:

Feature Name	Access Role	Enna Purpose & Epadi Use Aagum (Tanglish Summary)	Status
Daily Assessment Survey	Personnel	Sleep (0-12h), Stress slider (1-10), Heart rate, Social connection log panna.	**Active**
Triage Scoring Algorithm	System / All	Stress and sleep data vechu High (Red), Medium (Yellow), Low (Green) triage auto calculate aagum.	**Active**
4-7-8 Breathing Stabilizer	Personnel	Animated circle guide vechu 4s Inhale, 7s Hold, 8s Exhale seithu panic relief pera.	**Active**
Smartwatch Telemetry Sync	Personnel	1-click-la wearable sensor heart rate and sleep biometrics form-la load panna.	**Active**
1-Click Crisis SOS Beacon	Personnel / Officer	Army 1902 & Tele-MANAS 14416 call seiya + Officer screen-la pulsing red alert kaatta.	**Active**
Cumulative Fatigue Forecaster	Personnel / Officer	7-day sleep debt calculation panni Operational Strain Index (0-100%) and rest alert kaatta.	**Active**
Gamified Morale Badges	Personnel	Continuous self-logging seiyumbodhu daily streak and resilience badges unlock panna.	**Active**
Personnel Monitoring Roster	Officer	Command unit jawans list-ai triage color tags and filters vechu live monitor panna.	**Active**
Soldier Clinical Dossier	Officer	Oru soldier-oda complete historical trends, sleep graph, past notes review seiya.	**Active**
Confidential Counseling Notes	Officer	1-on-1 counseling mudithu findings, treatment plan, next review date secure-ah log panna.	**Active**
Executive Analytics Dashboard	Admin	Army-wide High, Medium, Low triage counts and unit health distribution overview.	**Active**
Direct Multi-Role Access	All Roles	PIN barrier illamal direct-ah Personnel, Officer, Admin onboarding seiyyum streamlined system.	**Active**
Unit Self-Service Recovery	All Roles	`/forgot_password` moolam Name + Service Unit submit panni password reset seiya.	**Active**
Immutable Activity Audit Trail	Admin / System	All critical events, logins, notes, PIN changes timestamp-udan log aagum.	**Active**
Vernacular Multi-Language	All Roles	Zero-latency-la English, Tamil, Hindi language switch seiya (`localStorage` persistence).	**Active**
Laptop 2-Col & Mobile Drawer UI	All Roles	Zero horizontal overflow, 2-column laptop grid, touch-friendly mobile drawer menu.	**Active**
Vercel Blob Cloud Persistence	Cloud Backend	Serverless cold boots-la SQLite database azhindhu pogamal automatic snapshot sync seiya.	**Active**
PDF Report Generators	Personnel/Officer/ Admin	Soldier Dossier (`/my/export_pdf`), Officer Report, Admin Command Report download panna.	**Active**
Delete Your Account	All Roles	Password verify panni user thanga account & biometric data-vai permanently erase seiya.	**Active**
Dedicated [■ Menu] Capsule	All Roles	Header capsule button, single authoritative click handler, elevated z-index 9999 smooth drawer.	**Active**
Zero-Emoji Vector SVGs	All Screens	Informal emojis removed, replaced with crisp high-DPI vector SVG icons across 24 templates.	**Active**
Defense & IT Dual-Use	All Workforces	Military patrol fatigue & IT sprint burnout, smartwatch band sync, and 24/7 AI wellness companion.	**Active**
Online PDF Viewers	Public / Browser	Direct in-browser PDF viewing via `/guide` (Features Guide) and `/documentation` (System Docs).	**Active**

Mudivu & System Guarantee: Indha APPS platform ippo military welfare monitoring-kku thevaiyaana anaithu modern tools-udan 100% production-ready aagi Vercel-la live-ah run aagudhu. Soldiers friction-free direct single-step login moolam entry podalaam, officers immediate triage action edukalaam, matrum Senior Admins command overview-ai securely maintain seiyalaam.